

TCS NQT Quick Reference

Aptitude Formulas, Shortcuts & Tips

TCS NQT Preparation Material
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1. Number System

Divisibility, HCF/LCM, remainders and properties of numbers.

Divisibility Rules

Divisor	Rule
2	Last digit is even (0, 2, 4, 6, 8)
3	Sum of digits divisible by 3
4	Last two digits form a number divisible by 4
5	Last digit is 0 or 5
6	Divisible by both 2 and 3
7	Double last digit, subtract from rest; result div by 7
8	Last three digits divisible by 8
9	Sum of digits divisible by 9
11	Difference of sum of alternate digits divisible by 11

HCF and LCM

Product Rule

$HCF(a, b) \times LCM(a, b) = a \times b$
Applies only for two numbers.

HCF of Fractions

$HCF = HCF(\text{numerators}) / LCM(\text{denominators})$

LCM of Fractions

$LCM = LCM(\text{numerators}) / HCF(\text{denominators})$

Tip: For three numbers: $LCM(a,b,c) = LCM(LCM(a,b), c)$

Remainder Theorems

Remainder Theorem

$(a \times b) \bmod n = [(a \bmod n) \times (b \bmod n)] \bmod n$

Fermat's Little Theorem

$a^{(p-1)} \bmod p = 1$, when p is prime and $\gcd(a,p)=1$

Wilson's Theorem

$(p-1)! \bmod p = p-1$, when p is prime

Tip: Cyclicity of units digits: 2- $\rightarrow$ {2,4,8,6}, 3- $\rightarrow$ {3,9,7,1}, 7- $\rightarrow$ {7,9,3,1}, 8- $\rightarrow$ {8,4,2,6}

2. Percentages

Conversion shortcuts and successive percentage changes.

Fraction-Percentage Conversion Table

Fraction	Percentage	Fraction	Percentage
1/2	50%	1/8	12.5%
1/3	33.33%	1/9	11.11%
1/4	25%	1/10	10%
1/5	20%	1/11	9.09%
1/6	16.67%	1/12	8.33%
1/7	14.28%	1/15	6.67%
2/3	66.67%	1/20	5%
3/4	75%	2/5	40%
4/5	80%	3/5	60%
5/6	83.33%	7/8	87.5%

Key Formulas

Percentage Change

$$\text{Change\%} = (\text{Difference} / \text{Original}) \times 100$$

Successive Percentage Change

$$\text{Net\%} = a + b + (a \times b)/100$$

For two successive changes of a% and b%.

Reverse Percentage

$$\text{If price increases by R\%, original} = \text{New} \times 100 / (100 + R)$$

Tip: To increase by 20% multiply by 1.2; to decrease by 20% multiply by 0.8.

3. Profit and Loss

Formulas and shortcut methods for profit/loss problems.

Profit %

$$\text{Profit\%} = (\text{Profit} / \text{CP}) \times 100$$

Loss %

$$\text{Loss\%} = (\text{Loss} / \text{CP}) \times 100$$

SP from Profit%

$$\text{SP} = \text{CP} \times (100 + \text{P\%}) / 100$$

SP from Loss%

$$\text{SP} = \text{CP} \times (100 - \text{L\%}) / 100$$

Discount

$$\text{SP} = \text{MP} \times (100 - \text{D\%}) / 100, \text{ where MP} = \text{Marked Price}$$

Two Successive Discounts

$$\text{Effective Discount} = a + b - (a \times b)/100$$

False Weight Gain%

$$\text{Gain\%} = (\text{True Weight} - \text{False Weight}) / \text{False Weight} \times 100$$

Tip: If CP of x items = SP of y items, then Profit% = $(x-y)/y \times 100$

Tip: Buy x get y free $\Rightarrow$ Discount% = $y/(x+y) \times 100$

4. Time, Speed and Distance

Relative speed, trains, boats, and circular motion.

Basic Formula

Distance = Speed $\times$ Time

Unit Conversion

1 km/hr = $\frac{5}{18}$ m/s | 1 m/s = $\frac{18}{5}$ km/hr

Average Speed (same distance)

Avg Speed = $2 \times S_1 \times S_2 / (S_1 + S_2)$
When equal distances are covered at two different speeds.

Trains

Train passing a pole/person

Time = Length of train / Speed of train

Train passing a platform

Time = (Length of train + Length of platform) / Speed

Two trains - same direction

Relative Speed = $|S_1 - S_2|$; Distance = Sum/Diff of lengths

Two trains - opposite direction

Relative Speed = $S_1 + S_2$; Distance = Sum of lengths

Boats and Streams

Downstream Speed

$D = B + S$ (boat speed + stream speed)

Upstream Speed

$$U = B - S$$

Still Water Speed

$$B = (D + U) / 2$$

Stream Speed

$$S = (D - U) / 2$$

Circular Motion

First Meeting (opposite direction)

$$\text{Time} = \text{Track Length} / (S_1 + S_2)$$

First Meeting (same direction)

$$\text{Time} = \text{Track Length} / |S_1 - S_2|$$

5. Time and Work

Work-rate problems, pipes and cisterns, efficiency method.

Basic Concept

If A can do work in 'n' days, A's 1 day work = $1/n$

Combined Work

$$1/T = 1/A + 1/B \Rightarrow T = (A \times B) / (A + B)$$

Three workers

$$1/T = 1/A + 1/B + 1/C$$

Wages Ratio

Wages are distributed in ratio of efficiencies (or work done)

Pipes and Cisterns

Filling pipe

Work rate = $+1/t$ (positive)

Emptying pipe (leak)

Work rate = $-1/t$ (negative)

Net rate

$$\text{Net} = 1/\text{Fill} - 1/\text{Empty}; \text{Time} = 1/\text{Net}$$

LCM / Efficiency Method

1. Take LCM of all given days as Total Work units.
2. Efficiency of each = Total Work / Individual days.
3. Combined efficiency = Sum of individual efficiencies.
4. Time = Total Work / Combined efficiency.

Tip: This method avoids fractions entirely -- use it in exams!

6. Averages and Mixtures

Weighted average, alligation rule.

Average

Average = Sum of observations / Number of observations

Weighted Average

$$W.Avg = (w_1 \cdot x_1 + w_2 \cdot x_2 + \dots) / (w_1 + w_2 + \dots)$$

New Average (adding a member)

$$\text{New Avg} = \text{Old Avg} + (\text{New value} - \text{Old Avg}) / (n + 1)$$

Alligation Rule

Alligation Formula

$$n_1/n_2 = (A_2 - \text{Avg}) / (\text{Avg} - A_1)$$

n_1, n_2 = quantities; A_1 = cheaper price; A_2 = dearer price; Avg = mean price

Tip: Draw the alligation cross: Cheaper -- Mean -- Dearer; differences give the ratio.

7. Ratio and Proportion

Proportion

If $a:b = c:d$ then $a*d = b*c$ (cross multiplication)

Componendo

$$(a+b)/b = (c+d)/d$$

Dividendo

$$(a-b)/b = (c-d)/d$$

Componendo-Dividendo

$$(a+b)/(a-b) = (c+d)/(c-d)$$

Direct Proportion

If x increases, y increases: $y = kx$

Inverse Proportion

If x increases, y decreases: $xy = k$

Tip: In mixture replacement: After n operations, remaining = $V \times (1 - R/V)^n$

8. Simple and Compound Interest

Simple Interest

SI Formula

$$SI = P \times R \times T / 100$$

Amount

$$A = P + SI = P(1 + RT/100)$$

Compound Interest

CI Formula

$$A = P(1 + R/100)^T$$

CI Amount

$$CI = A - P = P[(1 + R/100)^T - 1]$$

Half-yearly compounding

$$A = P(1 + R/200)^{(2T)}$$

Quarterly compounding

$$A = P(1 + R/400)^{(4T)}$$

Difference (2 years)

$$CI - SI = P(R/100)^2$$

Shortcut for 2-year problems.

Difference (3 years)

$$CI - SI = P(R/100)^2 \times (3 + R/100)$$

9. Permutations and Combinations

Factorial

$$n! = n \times (n-1) \times (n-2) \times \dots \times 1; 0! = 1$$

Permutation (nPr)

$$nPr = n! / (n-r)!$$

Arrangement of r items from n (order matters).

Combination (nCr)

$$nCr = n! / [r! \times (n-r)!]$$

Selection of r items from n (order does NOT matter).

Circular Permutation

$(n-1)!$ ways to arrange n items in a circle

With repetition

$$\text{Permutations} = n! / (p! \times q! \times \dots)$$

Where $p, q, \dots$ are frequencies of repeated items.

Key Identity

$$nCr = nC(n-r) \quad | \quad nC0 = nCn = 1$$

Pascal's Rule

$$nCr = (n-1)C(r-1) + (n-1)Cr$$

10. Probability

Basic Probability

$$P(E) = \text{Favourable outcomes} / \text{Total outcomes}$$

Complementary

$$P(E') = 1 - P(E)$$

Addition Rule

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Mutually Exclusive

$$P(A \text{ or } B) = P(A) + P(B) \text{ [when } P(A \text{ and } B) = 0]$$

Independent Events

$$P(A \text{ and } B) = P(A) \times P(B)$$

Conditional Probability

$$P(A|B) = P(A \text{ and } B) / P(B)$$

Dice: sum of two dice

$$\text{Total outcomes} = 36; P(\text{sum}=7) = 6/36 = 1/6 \text{ (highest)}$$

Cards

$$52 \text{ cards: } 4 \text{ suits} \times 13 \text{ ranks; } P(\text{Ace}) = 4/52 = 1/13$$

11. Geometry and Mensuration

Areas, volumes, surface areas of all common shapes.

2D Figures -- Area & Perimeter

Shape	Area	Perimeter
Square (side a)	a^2	$4a$
Rectangle (l x b)	$l \times b$	$2(l + b)$
Triangle (base b, ht h)	$(1/2) \times b \times h$	$a + b + c$
Equilateral Triangle (a)	$(\sqrt{3}/4) \times a^2$	$3a$
Circle (radius r)	$\pi \times r^2$	$2 \times \pi \times r$
Semicircle (r)	$(\pi \times r^2) / 2$	$\pi \times r + 2r$
Trapezium (a,b,h)	$(1/2)(a+b) \times h$	Sum of all sides
Parallelogram (b, h)	$b \times h$	$2(a + b)$
Rhombus (d1, d2)	$(1/2) \times d1 \times d2$	$4a$

3D Figures -- Volume & Surface Area

Shape	Volume	TSA / CSA
Cube (a)	a^3	TSA= $6a^2$, CSA= $4a^2$
Cuboid (l,b,h)	$l \times b \times h$	TSA= $2(lb+bh+lh)$, CSA= $2h(l+b)$
Cylinder (r, h)	$\pi \times r^2 \times h$	TSA= $2\pi r(r+h)$, CSA= $2\pi r \times h$
Cone (r, h, l)	$(1/3) \pi r^2 h$	TSA= $\pi r(r+l)$, CSA= $\pi r \times l$
Sphere (r)	$(4/3) \pi r^3$	$4 \pi r^2$
Hemisphere (r)	$(2/3) \pi r^3$	TSA= $3\pi r^2$, CSA= $2\pi r^2$

Important Geometry Facts

Heron's Formula

Area = $\sqrt{s(s-a)(s-b)(s-c)}$, where $s = (a+b+c)/2$

Angle sum of polygon

$(n-2) \times 180$ degrees, each interior angle of regular polygon = $(n-2) \times 180/n$

Arc and Sector

Arc length = $(\theta/360) \times 2\pi r$ | Sector area = $(\theta/360) \times \pi r^2$

12. Algebra Identities

Square of Sum

$$(a + b)^2 = a^2 + 2ab + b^2$$

Square of Difference

$$(a - b)^2 = a^2 - 2ab + b^2$$

Difference of Squares

$$a^2 - b^2 = (a + b)(a - b)$$

Cube of Sum

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

Cube of Difference

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

Sum of Cubes

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

Difference of Cubes

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Sum of first n naturals

$$n(n+1)/2$$

Sum of squares

$$n(n+1)(2n+1)/6$$

Sum of cubes

$$[n(n+1)/2]^2$$

AP: nth term

$$a_n = a + (n-1)d \quad | \quad \text{Sum} = n/2 [2a + (n-1)d]$$

GP: nth term

$$a_n = a \cdot r^{(n-1)} \mid \text{Sum} = a(r^n - 1)/(r - 1)$$

13. Shortcut Methods and Vedic Math Tricks

Quick Multiplication Tricks

Multiply by 11

Insert sum of digits between them: $23 \times 11 = 2_ (2+3) _ 3 = 253$
If sum > 9, carry over: $78 \times 11 = 7_ (7+8) _ 8 = 858$

Multiply by 5

Divide by 2 and multiply by 10: $48 \times 5 = 48/2 \times 10 = 240$

Multiply by 25

Divide by 4 and multiply by 100: $48 \times 25 = 48/4 \times 100 = 1200$

Multiply by 125

Divide by 8 and multiply by 1000

Multiply numbers near 100

97×96 : Deficiencies 3, 4. Answer: $(97-4)|(3 \times 4) = 93|12 = 9312$
Base method: $(a-d_2) \times 100 + d_1 \times d_2$ where d_1, d_2 are deficiencies from 100.

Quick Division Tricks

Divide by 5

Multiply by 2 and divide by 10

Divide by 25

Multiply by 4 and divide by 100

Divide by 50

Multiply by 2 and divide by 100

Common Pythagorean Triplets

Triplet	Multiple x2	Multiple x3
3, 4, 5	6, 8, 10	9, 12, 15
5, 12, 13	10, 24, 26	15, 36, 39
7, 24, 25	14, 48, 50	21, 72, 75
8, 15, 17	16, 30, 34	24, 45, 51
9, 40, 41	18, 80, 82	--
11, 60, 61	20, 21, 29	12, 35, 37

14. Squares and Cubes (1-30)

n	n ²	n ³
1	1	1
2	4	8
3	9	27
4	16	64
5	25	125
6	36	216
7	49	343
8	64	512
9	81	729
10	100	1000
11	121	1331
12	144	1728
13	169	2197
14	196	2744
15	225	3375

n	n ²	n ³
16	256	4096
17	289	4913
18	324	5832
19	361	6859
20	400	8000
21	441	9261
22	484	10648
23	529	12167
24	576	13824
25	625	15625
26	676	17576
27	729	19683
28	784	21952
29	841	24389
30	900	27000

15. Logical Reasoning Tips

Number Series Patterns

Common patterns to check:

1. Constant difference (AP): 2, 5, 8, 11, ... (diff = +3)
2. Increasing/decreasing differences: 1, 2, 4, 7, 11 (diff: 1,2,3,4)
3. Multiplication pattern: 2, 6, 18, 54 ($\times 3$)
4. Squares/Cubes: 1, 4, 9, 16, 25 or 1, 8, 27, 64
5. Alternating operations: +2, $\times 2$, +2, $\times 2$
6. Two interleaved series: 1, 10, 2, 20, 3, 30
7. n^2+1 or n^2-1 type: 2, 5, 10, 17, 26 (n^2+1)

Tip: Always compute first differences, then second differences. If second differences form a pattern, you have found it.

Coding-Decoding Methods

Letter coding: Check position shifts (A=1, B=2, ..., Z=26).

Reverse alphabet: A $\leftrightarrow$ Z, B $\leftrightarrow$ Y, C $\leftrightarrow$ X ... (position + reverse = 27)

Number coding: Map letters to numbers; check +1, +2, -1, reverse patterns.

Substitution coding: Real-world objects mapped to other words -- track consistently.

Tip: Make a quick A-Z numbered reference: A=1 through Z=26 before solving.

Blood Relations Shortcuts

Key: Draw a family tree diagram. Males on left, females on right.

Father's/Mother's son = Brother | daughter = Sister

Father's/Mother's brother = Uncle | sister = Aunt

Son's wife = Daughter-in-law | Daughter's husband = Son-in-law

Brother's/Sister's son = Nephew | daughter = Niece

Tip: Use + for male, - for female. Go generation by generation.

Direction Sense Tricks

Draw N-E-S-W cross. Start at origin.

Right turn from North = East, Left turn from North = West.

Opposite: N $\leftrightarrow$ S, E $\leftrightarrow$ W.

Use Pythagoras for final distance when path has right-angle turns.

Tip: Clockwise: N -> E -> S -> W. Counter-clockwise: N -> W -> S -> E.

Seating Arrangement Approach

1. Read ALL clues before drawing. Identify fixed positions first.
2. Use definite clues first (e.g., 'A sits at end'), then relative clues.
3. For circular: fix one person, place others relative to them.
4. For linear: mark Left-Right clearly; 'left of' means towards left end.
5. Use elimination -- if clue says 'not adjacent', rule out those spots.

Tip: Always verify your final arrangement against ALL given clues.

16. Mental Math Techniques

Squaring Numbers Near 50

Method

$$n^2 = (25 + d) \mid d^2, \text{ where } d = n - 50$$

Example: $53^2 \Rightarrow d=3$, Answer = $(25+3)|(3^2) = 28|09 = 2809$. $47^2 \Rightarrow d=-3$, Answer = $(25-3)|(3^2) = 22|09 = 2209$.

Squaring Numbers Near 100

Method

$$n^2 = (n + d) \mid d^2, \text{ where } d = n - 100$$

Example: $103^2 \Rightarrow d=3$, Answer = $(103+3)|(3^2) = 106|09 = 10609$. $97^2 \Rightarrow d=-3$, Answer = $(97-3)|(3^2) = 94|09 = 9409$.

Squaring Numbers Ending in 5

Method

$$n5^2 = n \times (n+1) \mid 25$$

Example: $35^2 = 3 \times 4 \mid 25 = 1225$. $85^2 = 8 \times 9 \mid 25 = 7225$.

Quick Percentage Calculations

10% of x: Move decimal one place left. 10% of 450 = 45.

5% of x: Half of 10%. 5% of 450 = 22.5.

15% of x: 10% + 5%. 15% of 450 = 45 + 22.5 = 67.5.

20% of x: Divide by 5. 20% of 450 = 90.

25% of x: Divide by 4. 25% of 450 = 112.5.

33.3% of x: Divide by 3. 33.3% of 450 = 150.

1% of x: Move decimal two places left. 1% of 450 = 4.5.

Tip: Combine: 17% = 10% + 5% + 2%. Build any percentage from 10%, 5%, 1%.

Estimation Techniques

1. **Round to nearest 10/100** for quick approximation.

2. **Use compatible numbers:** $498 \times 7 \sim 500 \times 7 = 3500$, adjust $-14 = 3486$.
3. **For fractions:** $29/31$ is close to 1; $7/15$ is close to $1/2$.
4. **Eliminate options:** Use units digit or divisibility to rule out wrong answers.
5. **Order of magnitude:** Quickly check if answer should be hundreds/thousands/lakhs.

Tip: In MCQs, estimation can save 30-40 seconds per question. Use it aggressively!

Quick Exam Tips

1. Attempt easy questions first -- scan the paper in first 2 minutes.
2. No negative marking in TCS NQT -- attempt ALL questions.
3. For calculation-heavy problems, use approximation and eliminate options.
4. Time per question: ~1 minute for aptitude. Skip if stuck > 90 seconds.
5. Practice mental math daily -- squares, cubes, fraction-percentage conversions.
6. For ratio problems, assume convenient values (LCM method).
7. Read the question carefully -- 'increased BY' vs 'increased TO' are different.
8. In geometry, always draw a rough figure.